HISTORICAL ASPECTS

Kawasaki syndrome (KS), originally called mucocutaneous lymph node syndrome, was first recognized and described in Japan in 1967 by Dr. Tomisaku Kawasaki. He reported his experience with 50 children who had been seen during the preceding 6-year period at the Tokyo Red Cross Medical Center. These children appeared to manifest a constellation of findings distinct from any previously described disease. After this description, numerous cases were reported throughout Japan. Dr. Kawasaki's first description of the disease in the English language appeared in 1974. In 1976, the syndrome was first reported in the United States by Melish, Hicks, and Larson in a group of 12 children from Honolulu seen between 1971 and 1973. Since then, KS has been recognized worldwide in children of every racial group. The Kawasaki Disease Research Committee published the fifth revised edition of its diagnostic guidelines in 2005.

EPIDEMIOLOGY

Although Kawasaki syndrome occurs worldwide in children of all racial groups, it is most prevalent in Japan and in children of Japanese ancestry. KS is primarily an illness of young children, with approximately 80% of patients younger than 5 years of age. The male-to-female ratio is about 1.5:1. The incidence rate is highest among Japanese children between 6 and 12 months of age, with similar numbers affected in the first and second years of life. Secondary cases rarely occur in the same school, daycare center, or household, suggesting that person-to-person transmission is unlikely. In Japan, the recurrence rate of KS is reported to be 3.9%, and the proportion of sibling cases is 1.4%. In patients whose parents had KS in childhood (0.2% of over 32,000 cases), there is a higher rate of sibling cases, increased likelihood of recurrence, greater need for intravenous immunoglobulin (IVIG), and more coronary abnormalities one month after onset. The rate of recurrence is highest within the first 12 months after the initial episode. These findings suggest a common exposure to an infectious agent in genetically predisposed individuals.

The incidence of KS in Japan increased from 1967 to the mid-1980s.

KAWASAKI SYNDROME

AT A GLANCE

Most common cause of acquired heart disease in children, with highest prevalence in those of Japanese ancestry.

Major Features of Acute Kawasaki Syndrome

- Fever
- Acute multisystem vasculitis
- Polymorphous exanthem
- Bilateral bulbar nonexudative conjunctival injection
- Inflammation of the oropharynx
- Extremity changes: erythema of the palms or soles, edema of the hands or feet, or periungual desquamation
- Acute non-suppurative cervical lymphadenopathy (≥1.5 cm in diameter)

Diagnostic Criteria for Kawasaki Syndrome

Fever lasting 5 or more days without another explanation, plus at least four of the following five criteria:

- Bilateral painless bulbar nonexudative conjunctival injection
- Changes in the oropharynx: injected or fissured lips, injected pharynx, or strawberry tongue
- Extremity changes: erythema of the palms or soles, edema of the hands or feet, or periungual desquamation

- Polymorphous exanthem
- Acute non-suppurative cervical lymphadenopathy ($\geq$1.5 cm in diameter)

> **Note:** A diagnosis of atypical Kawasaki syndrome may be made in patients with fever and fewer than four criteria if coronary artery aneurysms are present.

Complications of Kawasaki Syndrome

- Coronary artery abnormalities, including aneurysms
- Myocarditis
- Arthralgia and arthritis
- Urethritis
- Aseptic meningitis

Treatment

Prompt treatment with **intravenous immunoglobulin (IVIG)** and **aspirin** significantly reduces the risk of coronary artery abnormalities.